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# Statement of Purpose

I aim to pursue a PhD in Computer Science focused on wireless sensing. My decision to pursue graduate studies is a natural outcome of my experiences gained as a student at the Indian Institute of Technology, Kharagpur, working with Prof. Sandip Chakraborty in the UbiNet lab and my research internship at the University of Waterloo. My current research interest is in millimeter-wave (mmWave) sensing.

In truth, my background tells a slightly more complicated story. My initial interest was in embedded systems. I completed graduate-level courses and a semester project with Prof. **Arnab Sarkar** on scheduling multi-processor, multi-bus distributed real-time control systems. While I enjoyed the material, I realized much of embedded systems research is theoretical, and I feel more motivated by solving problems with direct real-world impact. My interest in sensing research stems from the same guiding motivation. I am particularly drawn to projects involving real-world actors, such as humans, vehicles, or robots, where the research can be deployed and its impact directly observed. Specifically, my interest in mmWave sensing is fueled by the vast range of applications it enables, which are made possible by Doppler processing and the physical properties of millimeter-wave. At the same time, the challenges of low angular resolution, sparsity, and multipath make mmWave sensing research both intellectually stimulating and rewarding. I have taken a somewhat exploratory trajectory through the initial years of my undergraduate career, and I hope that in graduate studies, I will focus on driving innovative, productive research.

In 2020, during my first year of college, I joined the Formula Student team as an electronics member and later became Electronics Lead, contributing to vehicle data collection, driver safety, and battery pack design. To resolve data logging issues caused by engine noise, I redesigned the system with modular data loggers connected via noise-resistant CAN lines, enabling real-time data on suspension travel, wheel speed, IMUs, and engine performance. I developed a plausibility circuit to perform safety checks on redundant acceleration pedal sensors during failures. For the battery pack, I researched extensively and optimized cell selection and design using a custom metric. Building a car from scratch taught me project management and interdisciplinary collaboration.

In the summer after my third year, I participated in the **MITACS Globalink Research Internship** program, working with Prof. **Philip Beesley** at the **University of Waterloo**. Our project involved creating a hanging installation that responded to environmental stimuli, including interactions with room occupants. A software-based 'digital twin' allowed emulated stimuli to trigger responses in the physical structure and vice versa. I developed distributed Node Controller boards for the structure connected using the MQTT protocol, enabling touch and motion sensing. I developed a part of the firmware and designed the Expansion Board PCBs to interface with the microcontroller. Through my interaction with Prof. Beesley and my peers participating in the program, I was exposed to the international perspective of higher education.

**Research on mmWave sensing**: Soon after that summer, I started working on mmWave sensing under the supervision of Prof. **Sandip Chakraborty**. Motivated to conduct research with real-world impact, I began developing an mmWave-based navigation aid for the visually impaired. To support this work, I was awarded the **CHANAKYA Fellowship** by the Department of Science and Technology, Government of India.

Estimating vehicle odometry using a single mmWave radar is under-explored [1–5]. Existing methods often yield coarse velocity estimates limited by Doppler resolution and struggle with dynamic environments [2, 6, 7], and those using deep learning make real-time processing challenging [5, 7, 8]. To address these issues, I developed a theoretical model of phase change in range-FFT peaks for accurate ego-speed estimation, which was accepted as a poster in **MobiSys**

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**2024** [9]. Subsequently, I created a high-resolution ego-speed estimation method that works reliably in dynamic settings without learning-based approaches. I also built a custom bot for experiments and collected over 10 km of mmWave, IMU, and Vicon data in indoor environments. This work led to a co-primary author manuscript under review as of December 2024.

Recently, I have been interested in Non-Line-of-Sight (NLOS) sensing using the mmWave radar. Methods in the current literature [10–15] do not allow the radar to "see" strictly at its back. As a part of my Master's thesis, I am exploring NLOS mmWave sensing by utilizing the backlobe of the patch antenna. I have already achieved preliminary results proving the feasibility of this modality.

At UIUC, I would like to work with Prof. **Deepak Vasisht** and Prof. **Elahé Soltanaghai**. I am intrigued by Prof. Vasisht's work on radar odometry [7, 16]. I wonder if we can use the Doppler velocities of static objects ($v_{ego}cos(\theta)$) to refine the angular resolution ($sin(\frac{\Delta\theta}{2}) = \frac{\Delta v_{d,res}}{2v_{ego}sin(\theta)}$), thus improving SLAM performance. His work on security in wireless [17, 18] is fascinating. I am curious whether RFProtect could be extended to sense and mask real humans by 'corrupting' corresponding frequency shifts. I am also excited by his recent research on pico-satellite communication [19–21]. I am deeply inspired by Prof. Soltanaghai's work on mmWave backscatter tags [22, 23] and tag uplink communication using Chirp Slope Shift Keying [24]. I plan to use a similar idea in my work to direct mmWave radiation toward the radar's backlobe and modulate them to distinguish them from normal reflections. Prof. Soltanaghai's work on using synthetic reflectors to explore occluded regions in cars [25] also closely aligns with my current work.

My long-term career goal is to become a professor in wireless sensing, leading a lab focused on interdisciplinary and *deployable* research that benefits society. Pursuing a PhD is a crucial step in this journey. The high quality of research at UIUC, reflected in frequent top-tier publications, and my alignment with Prof. Vasisht and Prof. Soltanaghai's work naturally make it my top choice. UIUC is an ideal place for realizing my goals, with its strong culture of innovation and entrepreneurship. UIUC's collaborative research culture will help me perform high-quality research with a broader impact and expose me to diverse perspectives. The strong graduate program in Computer Science at the University of Illinois Urbana-Champaign will provide me with opportunities and resources to perform state-of-the-art research in wireless sensing and create a significant impact.

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A statement of 1000 words

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